

Project Proposal: Speculative Power Sampling for Efficient Inference-Time Reasoning

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1 Motivation

Recent studies have begun to examine the nature of post-training, with works such as [arXiv:2504.13837](#) arguing that Reinforcement Learning (RL) does not genuinely incentivize novel reasoning capabilities in Large Language Models (LLMs), but rather acts as a mechanism to improve sample efficiency. While RL remains necessary for the overall alignment of models to human preferences and formatting, extracting and amplifying true reasoning power requires a different paradigm. Advancements in inference-time scaling, such as the *Power Sampling* method. By employing Markov Chain Monte Carlo (MCMC) techniques to sample from a sharpened target distribution p^α , models can achieve similar performance in complex mathematical and algorithmic reasoning tasks compared to model with RL, without additional training.

However, MCMC-based generation suffers from computational inefficiency. The iterative nature of evaluating and resampling candidates can incur an inference latency overhead of $> 8\times$, making it highly impractical real deployment. This project aims to solve the latency bottleneck of test-time compute scaling by introducing **Speculative Power Sampling**. We propose a novel framework that accelerates the MCMC sampling process, enabling LLMs to perform high-speed, System-2 mathematical and logical reasoning.

2 Initial Ideas / Proposed Research

Our core plan of attack is to merge the autoregressive MCMC generation of Power Sampling with the parallel-evaluation paradigm of Speculative Decoding.

Instead of relying on a large target model to autoregressively generate and iterate on candidate blocks, we introduce a lightweight **Draft Model** (e.g., a 0.6B parameter model) to act as the MCMC proposer. The draft model rapidly generates a block of B tokens (X_{new}) containing a Chain-of-Thought (CoT) reasoning step. The **Target Model** (e.g., a 7B parameter base model) then acts purely as a parallel evaluator, scoring the likelihood of the proposed block.

To ensure the final output strictly adheres to the sharpened reasoning distribution p_{large}^α , we derive a modified Metropolis-Hastings acceptance criterion. The transition probability A from the current block X_{old} to the proposed block X_{new} is defined as:

$$A = \min \left(1, \frac{P_{large}(X_{new})^\alpha \cdot P_{small}(X_{old})}{P_{large}(X_{old})^\alpha \cdot P_{small}(X_{new})} \right) \quad (1)$$

If the block is accepted, the reasoning chain proceeds. If rejected—indicating a potential logical flaw or calculation error—the target model does *not* fallback to autoregressive generation. Instead, the draft model simply resamples a new trajectory. This entirely decouples the generation bottleneck from the heavy target model, achieving high-speed, rigorous logical deduction.

3 Related Work

- **Limits of RL & Inference-Time Scaling:** Recent literature (e.g., [arXiv:2504.13837](#)) highlights that RL does not inherently teach reasoning. Conversely, *Reasoning with Sampling* (Karan & Du, 2025) proved that base models can achieve single-shot reasoning on par with RL by sampling from p^α . Our work bridges these insights by addressing the speed limitations of test-time scaling.

- **Speculative Decoding:** [Leviathan et al. \(2023\)](#) introduced speculative sampling to accelerate standard autoregressive decoding. We extend this by adapting the acceptance criteria for non-standard distributions (p^α) within an MCMC loop.

4 Evaluation

To determine the success and efficiency of our proposed method, we will evaluate the system across two main dimensions: **System Efficiency** and **Reasoning Performance**.

- **Metrics:** We will measure the *Acceleration Ratio* (tokens generated per second compared to the standard Power Sampling baseline), the *Acceptance Rate* of the target model under the p^α distribution, and the *Pass@1* accuracy.
- **Datasets:** We plan to evaluate on complex reasoning benchmarks such as a subset of **MATH500** (for multi-step mathematical derivations) and **HumanEval** (for algorithmic logic and code synthesis).

5 Feasibility / Implementation Plan

This project is highly feasible within the given timeframe.

Models & Hardware: We will utilize the Qwen3 model family, specifically **Qwen3-0.6B** as the Draft Model and **Qwen3-7B** as the Target Model. Sharing the same tokenizer and vocabulary significantly simplifies the engineering overhead of calculating joint probabilities. The experiments can be executed on a single A100 GPU on colab since the target model only performs parallel forward passes.

Implementation Focus: The primary engineering effort will be modifying the generation loop to implement the custom acceptance formula. A conceptual implementation of the core logic is outlined below:

```

1 # Draft model proposes a sequence block
2 draft_probs, x_new = draft_model.generate(context, block_size=B)
3
4 # Target model evaluates the proposed block in ONE parallel forward pass
5 target_logits = target_model.forward(concat(context, x_new))
6 target_probs = softmax(target_logits)
7
8 # Calculate acceptance ratio A based on the derived formula
9 # (Assuming p_large_old and p_small_old are cached from previous step)
10 ratio_large = (target_probs[x_new] ** alpha) / (target_probs[x_old] ** alpha)
11 ratio_small = draft_probs[x_old] / draft_probs[x_new]
12
13 A = min(1.0, ratio_large * ratio_small)
14
15 if random.uniform(0, 1) <= A:
16     accept(x_new)
17 else:
18     reject_and_resample() # Trigger draft model to propose again

```

Listing 1: Pseudocode for Speculative Power Sampling Acceptance Step